MSG No. 165.94 $P\bar{3}'c'1$ [Type III, trigonal]

Table 1: Wyckoff site: `2a`, site symmetry: `32.`

No.	position	mapping
1	$\left[0,\, 0,\, \frac{1}{4}\right]$	`[1,2,3,4,5,6]`
2	$\left[0,\, 0,\, \frac{3}{4}\right]$	`[7,8,9,10,11,12]`

Table 2: Wyckoff site: `2b`, site symmetry: `-3'..`

No.	position	mapping
1	$[0,\, 0,\, 0]$	`[1,2,3,7,8,9]`
2	$\left[0,\, 0,\, \frac{1}{2}\right]$	`[4,5,6,10,11,12]`

Table 3: Wyckoff site: `4c`, site symmetry: `3..`

No.	position	mapping
1	$[0,\, 0,\, z]$	`[1,2,3]`
2	$\left[0,\, 0,\, \frac{1}{2} - z\right]$	`[4,5,6]`
3	$[0,\, 0,\, -z]$	`[7,8,9]`
4	$\left[0,\, 0,\, z + \frac{1}{2}\right]$	`[10,11,12]`

Table 4: Wyckoff site: `4d`, site symmetry: `3..`

No.	position	mapping
1	$\left[\frac{1}{3},\, \frac{2}{3},\, z\right]$	`[1,2,3]`
2	$\left[\frac{2}{3},\, \frac{1}{3},\, \frac{1}{2} - z\right]$	`[4,5,6]`
3	$\left[\frac{2}{3},\, \frac{1}{3},\, -z\right]$	`[7,8,9]`
4	$\left[\frac{1}{3},\, \frac{2}{3},\, z + \frac{1}{2}\right]$	`[10,11,12]`

Table 5: Wyckoff site: `6e`, site symmetry: `-1'`

No.	position	mapping
1	$\left[\frac{1}{2},\, 0,\, 0\right]$	`[1,7]`
2	$\left[0,\, \frac{1}{2},\, 0\right]$	`[2,8]`
3	$\left[\frac{1}{2},\, \frac{1}{2},\, 0\right]$	`[3,9]`
4	$\left[\frac{1}{2},\, 0,\, \frac{1}{2}\right]$	`[4,10]`
5	$\left[0,\, \frac{1}{2},\, \frac{1}{2}\right]$	`[5,11]`
6	$\left[\frac{1}{2},\, \frac{1}{2},\, \frac{1}{2}\right]$	`[6,12]`

Table 6: Wyckoff site: `6f`, site symmetry: `.2.`

No.	position	mapping
1	$\left[x,\, 0,\, \frac{1}{4}\right]$	`[1,4]`

continued ...

Table 6

No.	position	mapping
2	$\left[0,\, x,\, \frac{1}{4}\right]$	[2,5]
3	$\left[-x,\, -x,\, \frac{1}{4}\right]$	[3,6]
4	$\left[-x,\, 0,\, \frac{3}{4}\right]$	[7,10]
5	$\left[0,\, -x,\, \frac{3}{4}\right]$	[8,11]
6	$\left[x,\, x,\, \frac{3}{4}\right]$	[9,12]

Table 7: Wyckoff site: 12g, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$[-y,\, x-y,\, z]$	[2]
3	$[-x+y,\, -x,\, z]$	[3]
4	$\left[x-y,\, -y,\, \frac{1}{2}-z\right]$	[4]
5	$\left[y,\, x,\, \frac{1}{2}-z\right]$	[5]
6	$\left[-x,\, -x+y,\, \frac{1}{2}-z\right]$	[6]
7	$[-x,\, -y,\, -z]$	[7]
8	$[y,\, -x+y,\, -z]$	[8]
9	$[x-y,\, x,\, -z]$	[9]
10	$\left[-x+y,\, y,\, z+\frac{1}{2}\right]$	[10]
11	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	[11]
12	$\left[x,\, x-y,\, z+\frac{1}{2}\right]$	[12]